

RESEARCH ARTICLE

WILEY

Increasing plant diversity and forb ratio during the revegetation processes of trampled areas and trails enhances soil infiltration

Lei Su^{1*} | Yongsheng Yang^{2*} | Xiaoya Li¹ | Dong Wang¹  | Yanchun Liu¹ | Yinzhan Liu¹ | Zhongling Yang¹ | Mingming Li³

¹School of Life Sciences, Henan University, Kaifeng, Henan 475004, PR China

²Northwest Institute of Plateau Biology, Chinese Academy of Sciences, Xining 810001, PR China

³College of Environment and Planning, Henan University, Kaifeng, Henan 475004, PR China

Correspondence

Dong Wang, School of Life Sciences, Henan University, Kaifeng, Henan 475004, China.
Email: wangdong19882005@163.com

Funding information

National Natural Science Foundation of China, Grant/Award Numbers: 31600380, 41807158, 31600379, 31670477 and 31570429; National Science Foundation for Post-doctoral of China, Grant/Award Numbers: 2016M592284 and 2018T110722

Abstract

Livestock breeding and production activities provide the livelihood for herdsmen; nevertheless, the accompanying trampling has the potential to negatively influence the physiochemical environment of soil. The policy of returning grazing land to grassland has greatly influenced both biodiversity and ecosystem services. However, the regulatory mechanism behind changes of soil infiltration capacity remains largely unexplored. Here, three sites (a 1-year [R1] and a 3-year [R3] restoration of grassland, which were both trampled areas and trails an untrampled grassland) were sampled on the Mongolian Plateau of China. The infiltration parameters, soil properties, root channel area, and plant community were determined for each site. The results showed that the species richness, aboveground biomass, and the proportion of forbs in the R3 grassland were 2.7-, 5.5- and 30.8-times higher than those of the R1 grassland, respectively. The soil infiltration rates and cumulative infiltration increased with species richness, aboveground biomass, and the proportion of forbs. Importantly, all values of species richness, aboveground biomass, and forbs ratio increased with progressing restoration. Overall, our results indicate that grasses recovered first after trampling ceased. Along with the improvement of soil surface conditions, forb species with greater root channel area increased gradually, which increased the soil infiltration rate. Our research highlights the changes in plant community of trampled areas and trails and the regulatory influence of revegetation on soil infiltration.

KEYWORDS

infiltration rate, natural restoration, plant community, temperate steppe, trampled areas and trails

1 | INTRODUCTION

With increasing livestock and production activities on grassland, trampling intensifies. The impacts, almost regardless of their form, are often negative, tend to cause destruction of the natural soil structure, and cause undesirable grassland change (Liu & Wang, 2014; Pickering & Growcock, 2009). Many studies have reported severe consequences of grassland trampling. Trampling-induced vegetation loss enhances the temperature of the soil surface, which

can subsequently stimulate soil evaporation of the trampled areas and trails (Roundy, Winkel, Khalifa, & Matthias, 1992). Furthermore, the damaged soil structure may cause land degradation and is unfavourable for soil water retention (Ball, Cameron, Di, & Moore, 2012; Pietola, Horn, & Yli-Halla, 2005; Taboada & Lavado, 1993). These effects may lead to decreased soil hydraulic conductivity and increased run-off and thus constrain rainwater from permeating into deep soil and aggravate soil erosion (Bagarello, Iovino, & Elrick, 2004; Dunne, Western, & Dietrich, 2011; Herrick, Van Zee, Belnap, Johansen, & Remmenga, 2010; Li, Gao, & Tang, 2016). Soil erosion has become an inevitable topic of policy formulation (Yan, Fang,

*These authors contributed equally to this paper.

Zhang, & Shi, 2013). Currently, the Chinese government has gradually realized the prominent value of the grassland biome for combating land degradation and ensuring sustainable human development. Returning grazing land to grassland has led to large-scale grassland restoration in arid and semi-arid areas (Miao et al., 2015; Shao et al., 2016).

In water-stressed areas, soil water is a key limiting resource for plant growth and reproduction, and the available water greatly determines both ecosystem productivity and ecological functions (Rodríguez-Iturbe & Porporato, 2005; Zhao, Wu, Zhao, & Feng, 2013). Rainwater infiltration into soil is a pivotal process for replenishing the root-zone soil moisture, and the capacity of rainfall migrating into the soil is greatly controlled by the infiltration rate (Archer et al., 2016; Mao et al., 2016). Infiltration is particularly important for modulating ecohydrological processes in arid and semi-arid areas where rainfall is the only source of soil water replenishment and land remediation (Huang, Tian, Wu, Liu, & Dang, 2017; Kato, Onda, Tanaka, & Asano, 2009; Razzaghi, Khodaverdilo, & Dashtaki, 2016). Plant traits and soil structure are the two central determinants of the soil infiltration capacity (Agnese, Bagarello, Baiamonte, & Iovino, 2011; Fischer et al., 2015; Leung, Garg, Co, Ng, & Hau, 2015). Roots are a critical factor that influences soil pore geometry (Milleret, Le Bayon, Lamy, Gobat, & Boivin, 2009). Wu, et al. (2016) reported that artificial grasslands with more root biomass have higher infiltration capacity. The plant root system can generate biopores named root channels (Jørgensen et al., 2002; Wu et al., 2017). Root channels produce soil macroporosity, which causes lateral rainwater flow and rapidly replenishes groundwater during rain events (Weiler & Naef, 2003; Lange, Lüescher, & Germann, 2009; Stumpf, Pauletto, & Pinto, 2016).

Both plant community and soil properties are not static during the natural restoration processes of grassland (Breuer, Huisman, Keller, & Frede, 2006; Critchley et al., 2002; Miao, Qiu, Guo, Musa, & Jiang, 2018). Plants begin to shoot up after the elimination of trampling interference, and a large number of plants settle down. Their roots extend gradually, soil properties change correspondingly, and the establishment of vegetation and the accompanied alternation of soil structure contribute to changes of soil hydrological processes (Clark & Zipper, 2016; Fields-Johnson, Zipper, Burger, & Evans, 2012). Therefore, it is necessary to investigate how the soil infiltration capacity changes during the conversion from trampled areas and trails to grassland.

The objective of the present study is to study the regulatory effect of changes in the plant community during the natural conversion processes of trampled areas and trails to grassland on the infiltration processes. The soil properties, plant community, soil infiltration ability of both a 1-year (R1) and a 3-year (R3) restoration grassland, which were recovered from trampled areas and trails, and a protected grassland were measured.

2 | MATERIALS AND METHODS

2.1 | Site description

This study was performed in a typical temperate steppe, which is located at Duolun County (42°02'N, 116°17'E, 1324 m a.s.l.) of the Inner Mongolia Autonomous Region, China. The long-term

(1954–2017) mean annual precipitation of the area is 382.2 mm, approximately 90% of annual rainfall falls during the growing season (May to October), whereas the non-growing season is dry and cool. The mean annual air temperature is 2.4°C; the mean monthly maximum temperature of 19.1°C occurs in July, whereas January is the coldest month with an average monthly temperature of −17.3°C (China Meteorological Data Sharing Service System). The plant community is dominated by the following perennial species: *Stipa krylovii*, *Agropyron cristatum*, *Cleistogenes squarrosa*, *Artemisia frigida*, *Potentilla acaulis*, and *Allium bidentatum*.

2.2 | Experimental design

The study objectives were a North–South trending trackway (including one half suffered trampling until 1 year ago and the other half began to recover 3 years ago) and adjacent grassland (Figure 1). The R1 grassland was trampled by humans during 2013 and 2016. The grassland adjacent to it was a R3 grassland, which was compacted by trampling from 2011 to 2014; the trampling intensity was identical to that of R1 and had reached the recovery phase since 2014. The control grassland (C) was adjacent to the restoration grassland, which was enclosed since 2000 and remained without interference.

Four blocks were selected in August 2017, and the length and width of each block were 20 and 6 m, respectively. To minimize spatial heterogeneity, three plots along the north–south direction were selected in each block, and the distance between any two plots was <5 m (Gao, Shao, & Wang, 2012; Khodaverdilo et al., 2017). Three 1 m × 1 m subplots were assigned to C, R1, and R3 in each plot. The experiment included 12 plots and 36 subplots (Figure 1). The mean values of the three subplots were considered as the values of the plots.

2.3 | Plant and soil property measurements

In each subplot, all aboveground plant parts were harvested, and the collected materials were sorted into envelopes by species. Then, the root channel of the stubble on the ground and soil infiltration process were measured, followed by soil and root samplings. A 9-cm-diameter root auger was employed to collect soil samples of 0–10 and 10–20 cm depth levels, and a 2-mm sieve was used to separate plant roots from soil. The aboveground plant parts and separated roots were oven-dried at 75°C for 48 hr and weighted to determine aboveground biomass and root biomass, respectively. Following common practice, the plants in the study plots were classified into two functional groups: grass and forb.

Surface soil samples at depths of 0–10 and 10–20 cm were selected. Soil cores were taken (using cutting-ring) for laboratory assays of soil moisture and bulk density (BD). The soil aggregates were immersed in distilled water and subsequently stirred by moving the aggregates up and down 60-times using a 0.2-mm sieve by a soil aggregate analyzer (DIK-2001). The soil capillary water content (W_c) was determined with the cutting-ring method. The undisturbed soil samples were sampled with cutting ring and transported to the laboratory. The upper cover was removed, and the lower cover with holes was immersed in water for 12 hr, with the water being 2 mm lower than the cutting ring. The lower cover was removed, and the cutting ring was placed on the plate

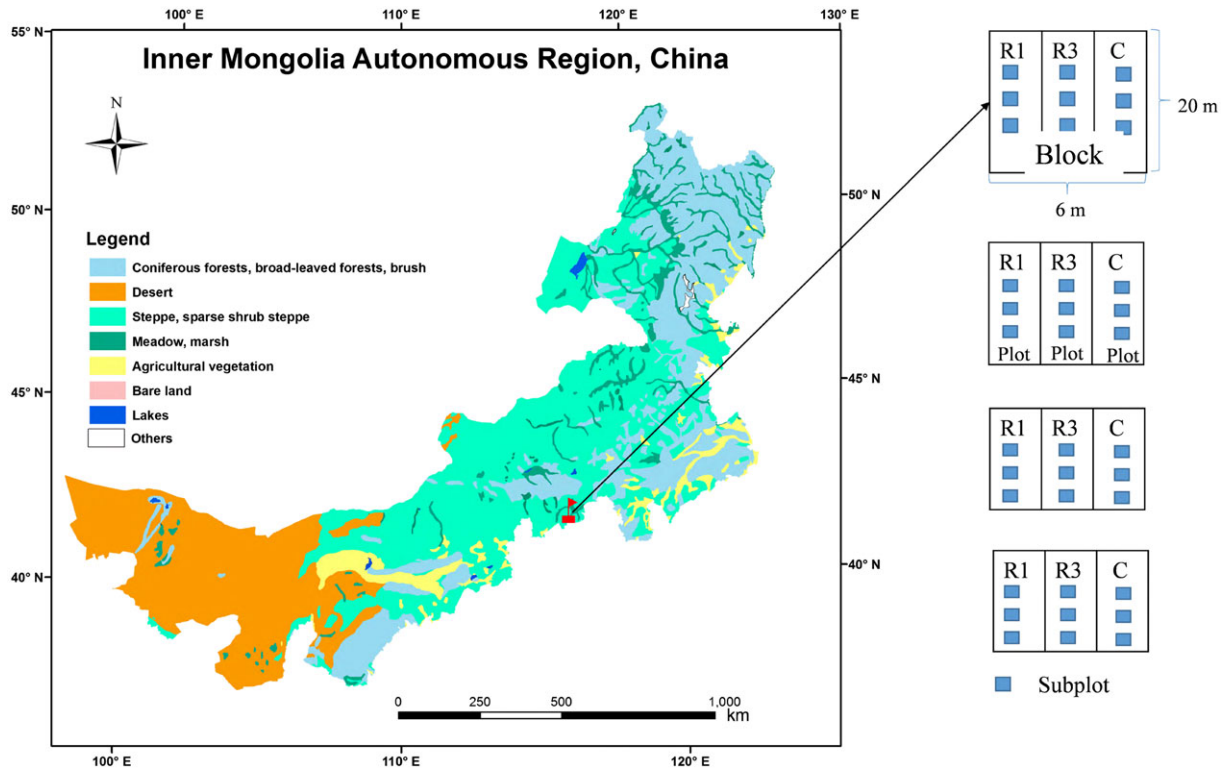


FIGURE 1 Location of the study site and arrangement of the sample plot [Colour figure can be viewed at wileyonlinelibrary.com]

equipped with dry soil for 2 hr; the weight at this point was m_1 (g). Then, they were oven-dried at 105°C and weighted; the gained weight was m_2 (g). The soil capillary water content was calculated as

$$W_c = \frac{m_1 - m_2}{m_2 - m_0} \cdot 100, \quad (1)$$

where W_c represents the content of soil capillary water (%) and m_0 represents the weight of cutting ring (g).

Soil capillary porosity (CP) was estimated using the formula used by Huang et al. (2017):

$$CP = W_c \cdot \frac{BD}{V} \cdot 100, \quad (2)$$

where CP represents the soil capillary porosity (%) and V represents the volume of soil core (cm^3).

2.4 | Root channel measurement

A vernier caliper was employed to measure root channel diameters at the soil surface. The average root channel diameter (ARCD) of each subplot was calculated by averaging the four parts of the subplot. The root channel area (RCA) of a subplot was derived from the total amount of root channels in the subplot. The derivation procedures of ARCD and RCA were expressed using the following equations:

$$ARCD = \frac{\sum_{i=1}^m d_i}{m(m=1, 2, 3, \dots)} \quad (3)$$

$$RCA = \frac{M \cdot \pi \cdot ARCD^2}{4S(M=1, 2, 3, \dots)}, \quad (4)$$

where m represents the amount of root channels in 1/4 of the subplot (mm^2), d represents the diameter of root channels (mm), S represents

the area of a subplot ($1 \text{ m}^2 = 10,000 \text{ cm}^2$), and M represents the amount of root channels in the subplot.

2.5 | Infiltration measurement

The infiltration process was determined via the point source method, which is based on the relationship between soil infiltrability and the advancement of water at the soil surface (Mao et al., 2016). Prior to measurement, the standing vegetation was mowed, and all litter was cleaned up from the soil surface. The experimental device consisted of a peristaltic pump, a peristaltic pump silicone tube, a camera, a tripod, and a computer. The peristaltic pump and assorted peristaltic pump silicone tube were applied to provide a constant water flow (3 L hr^{-1}) at the centre of the measured soil surface. The camera was fixed with the tripod, and the computer installed with a specific software and connected to the camera to obtain relevant data. The experimental device and peristaltic pump started to work simultaneously. The camera recorded the profiles of the expanding wetted soil surface every 3 min, and the measuring processes lasted for 60 min. Mathematical models were applied to compute the soil infiltrability from the recorded changed area of the wetted soil surface. The infiltration rates at specific time points were derived via the following equation:

$$i_n = \frac{q - \sum_{j=1}^{n-1} \Delta A_{n-j+1}}{\Delta A_n (n=1, 2, 3, \dots)} \quad (5)$$

where i_n represents the soil infiltration rate at time n (mm hr^{-1}), q represents the water flow rate (L hr^{-1}), in this study, $q = 3 \text{ L hr}^{-1}$, and ΔA_n represents the corresponding increased wetted area at the soil surface within time t_n and t_{n-1} (mm^2).

The infiltration process was divided into three different phases according to its dynamic process (Wu, et al., 2016). At the initial

stage of this process (0–15 min), the infiltration rate decreased rapidly; this process formed the fast infiltration phase, and the average infiltration rate of 0–15 min was considered as the infiltration rate of this phase (IRI). Subsequently, the decrease of infiltration rate became gentle during 15–45 min; this phase was named the stable change phase, whereas the average infiltration rate of 15–45 min was defined as the infiltration rate of phase II (IRII). The infiltration rate of 45–60 min remained stable, and the average infiltration rate of 45–60 min was defined as the third phase of the infiltration process (IRIII).

The cumulative infiltration can be calculated as

$$I_c = \sum I R_i \cdot t_i, \quad (6)$$

where I_c represents the cumulative infiltration (mm), subscript i represents the number of measurement times, and $I R_i$ and t_i represent the mean infiltration rate (mm hr^{-1}) and length (h) of measurement time i , respectively.

2.6 | Statistical analysis

SPSS 16.0 software was employed to statistically analyse all data. The differences in soil characteristics, vegetation indexes, and infiltration rate of grassland under different treatments were examined via one-way analysis of variance. The relativity between infiltration rates and soil properties or vegetation indexes was accessed via correlation analysis. All data were expressed as mean values $\pm$ standard error, and the significant differences were evaluated at the 0.05 probability level.

3 | RESULTS

3.1 | Infiltration rate and cumulative infiltration

The IRI was highest in C grassland ($88.7 \pm 5.8 \text{ mm hr}^{-1}$) and decreased by 27% and 31% in R3 grassland ($64.5 \pm 7.3 \text{ mm hr}^{-1}$) and R1 grassland ($61.1 \pm 3.0 \text{ mm hr}^{-1}$), respectively. The highest IRII was found in C grassland ($33.1 \pm 1.8 \text{ mm hr}^{-1}$), followed by R3 grassland ($23.0 \pm 2.6 \text{ mm hr}^{-1}$), whereas the lowest IRII was observed in R1 grassland ($16.2 \pm 1.1 \text{ mm hr}^{-1}$). The changes in IRIII were similar to

those observed for IRII (Figure 2a). The cumulative infiltration was highest in C grassland, which averaged $44.7 \pm 2.5 \text{ mm}$ in the range of 33.8–59.7 mm, followed by R3 grassland ($32.1 \pm 3.4 \text{ mm}$). The lowest cumulative infiltration was found in R1 grassland ($26.2 \pm 1.3 \text{ mm}$; Figure 2b).

3.2 | Soil properties

At the depth of 0–10 cm, the highest bulk density was found in R1 grassland, followed by R3 grassland, whereas the lowest bulk density was found in the C grassland. Soil moisture showed the opposite tendency with bulk density: It was highest in C grassland and lowest in R1 grassland, and the value of R3 grassland was in between. However, at the depth of 10–20 cm, the differences in bulk density and soil moisture among C grassland, R1 grassland, and R3 grassland were not significant (Figure 3; $p > 0.05$; Table 1). There was no significant difference in soil aggregate among the three grasslands both at the depths of 0–10 and 10–20 cm. The capillary porosity was highest in C grassland, decreased by 30% and 24% in R1 grassland and R3 grassland at the depth of 0–10 cm and decreased by 33% and 24% in R1 grassland and R3 grassland at the depth of 10–20 cm, respectively. The root biomass of C grassland and R3 grassland was higher than that of R1 grassland at both 0–10 and 10–20 cm. The root channel area was lowest in R1 grassland and increased by 22-fold and 17-fold in C and R3 grasslands, respectively (Figure 3; Table 1).

3.3 | Aboveground biomass and species richness

The mean aboveground biomass of C grassland was $165.2 \pm 4.7 \text{ g m}^{-2}$, ranging from 129.5 to 195.1 g m^{-2} , and averaged $21.3 \pm 2.9 \text{ g m}^{-2}$, ranging from 10.9 to 45.5 g m^{-2} for the R1 grassland, whereas the mean biomass was $118.0 \pm 8.1 \text{ g m}^{-2}$ in the range of 67.7–165.1 g m^{-2} for R3 grassland (Figure 4a). Forbs accounted for 58.8% of total biomass in C grassland but only supplied 1.1% of total biomass in the R1 grassland; their share increased to 33.9% in the R3 grassland (Figure 4b). The species richness of R1 grassland averaged 2.5 ranging from 2 to 3. It increased by 170% and reached 6.8 in the range of 4–9 for the R3 grassland.

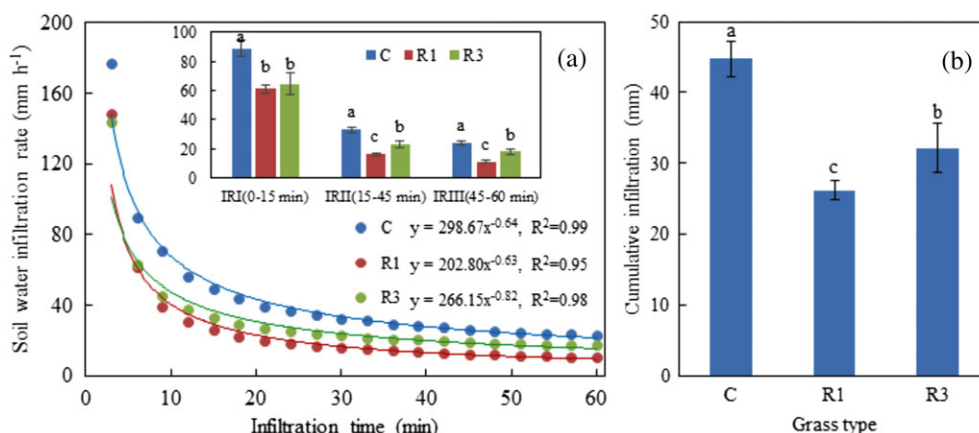


FIGURE 2 Soil infiltration parameters under different grasslands. Values with different letters in the same phases were significantly different at 0.05 level [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.3173)]

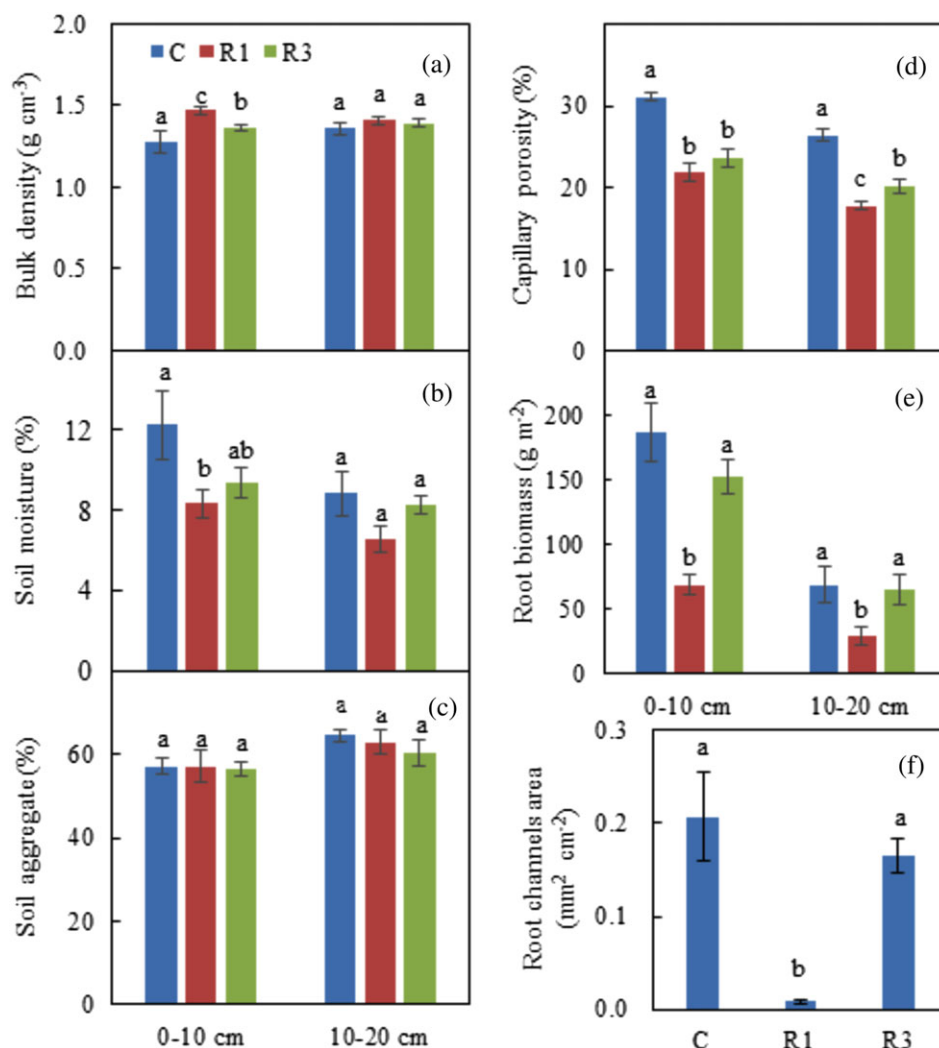


FIGURE 3 Soil properties, root biomass, and root channel area of the three grasslands. Values with different letters in the same columns were significantly different at 0.05 level [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.3173)]

TABLE 1 *F* values and *p* values of ANOVA comparing the soil properties, root channel characteristics, and infiltration parameters among different grasslands

Response variable	cm	<i>F</i> value	<i>p</i> value
Bulk density	0-10	21.79	<0.01
	10-20	1.23	0.31
Soil aggregate	0-10	0.03	0.97
	10-20	0.65	0.53
Total porosity	0-10	21.79	<0.01
	10-20	1.23	0.31
Capillary porosity	0-10	27.32	<0.01
	10-20	34.15	<0.01
Soil moisture	0-10	3.14	0.06
	10-20	2.46	0.10
Root biomass	0-10	14.36	<0.01
	10-20	3.63	0.04
Root channel area		12.43	<0.01
IRI (0-15 min)		7.12	0.003
IRII (30-45 min)		19.81	<0.01
IRIII (45-60 min)		21.22	<0.01
Cumulative infiltration		13.554	<0.01

Note. ANOVA: analysis of variance; IR: infiltration rate.

3.4 | Root diameter of different species

The root diameter at the depth of 10 cm of the 15 forb species averaged 0.68 ± 0.05 mm in the range of 0.46–1.22 mm, which was two-fold of the seven grass species; the root diameter at the depth of 10 cm of grass species ranged from 0.26 to 0.47 mm with a mean of 0.34 ± 0.04 mm (Figure 5).

3.5 | Correlations between plant community, soil properties, and infiltration parameters

The infiltration rates of IRI, IRII, and IRIII as well as cumulative infiltration were significantly positively correlated to soil moisture, species richness, aboveground biomass, and the proportion of forb functional group (Figure 6; all $p < 0.05$).

The infiltration rates of IRI, IRII, and IRIII as well as the cumulative infiltration were significantly positively associated with capillary porosity at the depth of 0–10 and 10–20 cm (all $p < 0.05$; Table 2). The values of IRII, IRIII, and cumulative infiltration were significantly negatively correlated to bulk density at the depth of 0–10 cm (all $p < 0.01$). Soil aggregate was not correlated to infiltration rates or

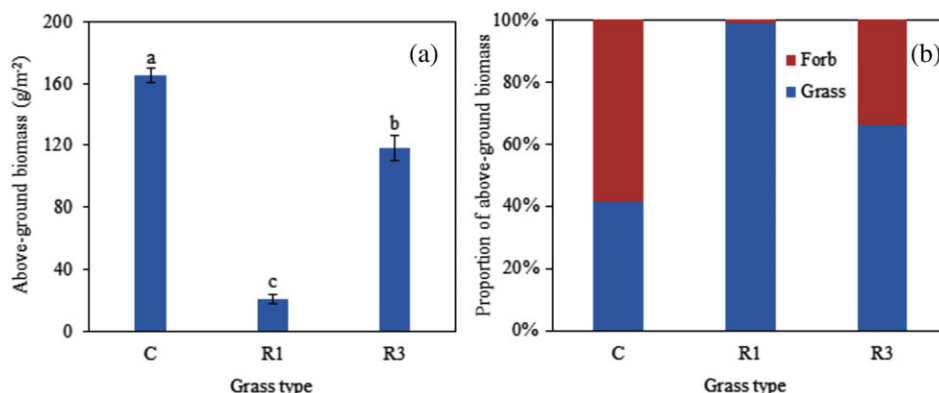


FIGURE 4 (a) Aboveground biomass and (b) ratio of functional group to aboveground biomass of the three grasslands. Values with different letters were significantly different at 0.05 level [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

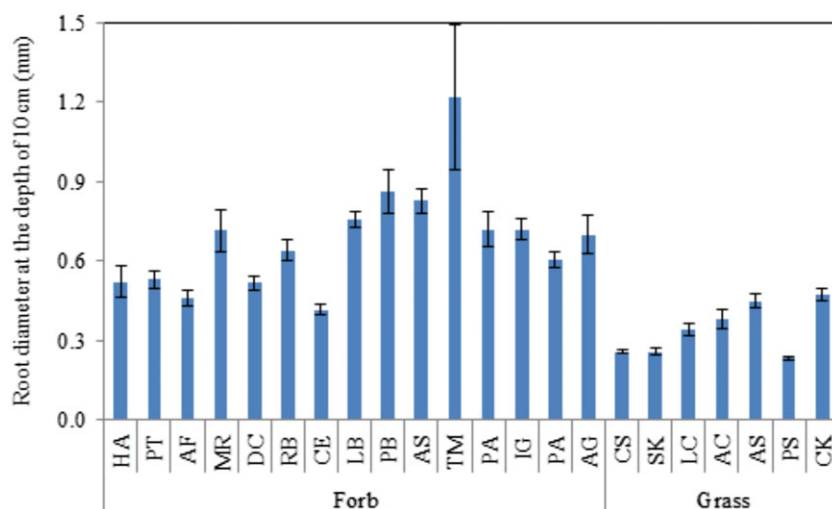


FIGURE 5 Root diameter at the depth of 10 cm of the main species in the natural grassland. CS: *Cleistogenes squarrosa*; SK: *Stipa krylovii*; LC: *Leymus chinensis*; AC: *Agropyron cristatum*; AS: *Achnatherum sibiricum*; PS: *Poa subfastigiata*; CK: *Carex korshinskyi*; HA: *Heteropappus altaicus*; PT: *Potentilla tanacetifolia*; AF: *Artemisia frigida*; MR: *Melissilil ruthenicus*; DC: *Dianthus chinensis*; RB: *Radix bupleuri*; CE: *Chamaerhodos erecta*; LB: *Lespedeza bicolor*; PB: *Potentilla bifurca*; AS: *Astragalus scaberrimus*; TM: *Taraxacum mongolicum*; PA: *Potentilla acaulis*; IG: *Ixeridium graminifolium*; PA: *Potentilla anserine*; AG: *Astragalus galactites* [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

cumulative infiltration (all $p > 0.05$). The root channel area was not correlated to IRI, IRII, or cumulative infiltration (all $p > 0.05$); however, it was significantly positively correlated to IRIII ($p < 0.05$).

4 | DISCUSSION

Previous studies have indicated that the soil permeability was largely determined by the physical properties of soil; soil infiltration responded significantly negatively to bulk density, whereas it correlated significantly and positively to capillary porosity (Sun et al., 2018; Yang & Zhang, 2011). As shown in this study, converting trampled areas and trails to grassland had a positive effect on the soil physical properties, such as decreasing the bulk density and enhancing the capillary porosity of the upper soil layer (Figure 3a,d), which may promote water transport (Blanco-Canqui, Wienhold, Jin, Schmer, & Kibet, 2017). This finding was verified by previous studies regarding the influence of vegetation on soil infiltration. For example, Jiménez, Tejedor, Morillas, and Neris (2006) found that harvesting natural vegetation can decrease soil infiltration because the removal of vegetation would lead to an increase in bulk density and a decrease in porosity.

Accompanying the removal of trampling, the compaction forces disappear, and the upper soil layer becomes less dense. Lower bulk

density is beneficial to support the growth of the root system. Plant roots release exudate, enmesh, and realign soil particles, which would lead to changes in soil through forming well-connected channels and macropores (Wu et al., 2017). Moreover, with the progressing restoration process, the accumulation of organic matter (such as carbohydrates, lignins, cellulose, adipose, and proteins) would promote the activities of soil animals, which may create a large number of soil pores and lessen the upper soil layer, therefore contributing to the development of soil capillary porosity (Luna, Vignozzi, Miralles, & Solé-Benet, 2018; Wu, Liu, Fang, Deng, & Shi, 2016; Zhao et al., 2013).

The soil water infiltration capacity is deeply influenced by the composition of the plant community (Fischer et al., 2015; Thompson, Harman, Heine, & Katul, 2010; Wu, Zhang, Wang, Shi, & Zhu, 2014), whereas the plant community succession is greatly controlled by the abiotic environment (Frouz et al., 2008; Wang, Zhang, Zhu, Yang, & Li, 2018). During the early period of the recovery processes of trampled areas and trails, grasses occupy the vast majority of the plant community. The grasses are a more resource-acquisitive plant functional group with thinner and shallower rooted species (Stampfli, Bloor, Fischer, & Zeiter, 2018). Consequently, they are able to quickly re-sprout from basal meristematic tissue and pre-empt space and nutrients (Volaire, Barkaoui, & Norton, 2014). These traits would help grasses to survive once trampling has been eliminated. Nevertheless,

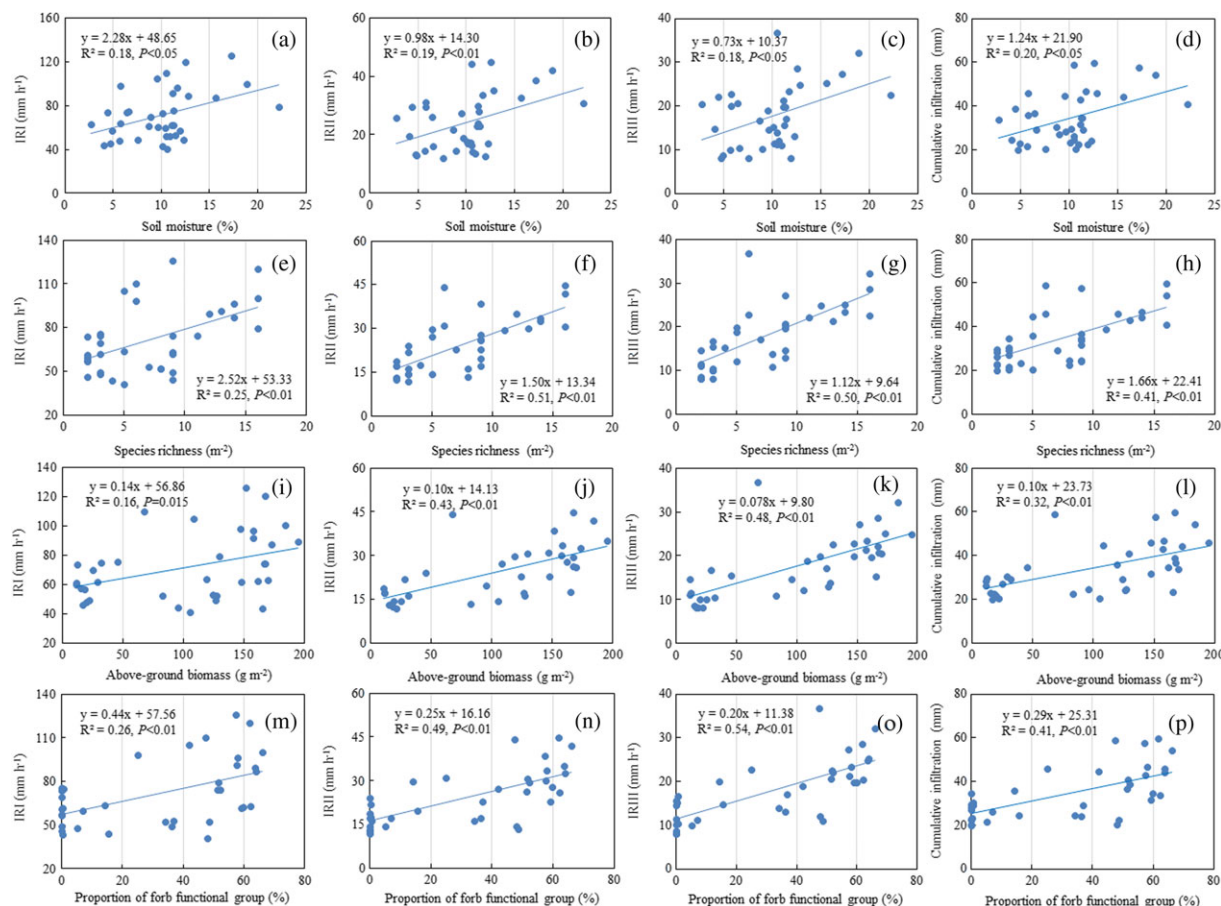


FIGURE 6 Correlations between soil moisture, species richness, aboveground biomass, forb functional group ratio, and infiltration parameters [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Correlation matrix (Pearson correlation coefficient) of bulk density, capillary porosity, soil aggregate, root channel area, and infiltration parameters

	Bulk density		Capillary porosity		Soil aggregate		Root channel area (mm ² cm ⁻²)
	0–10 cm	10–20 cm	0–10 cm	10–20 cm	0–10 cm	10–20 cm	
IRI (mm hr ⁻¹)	-0.226	-0.09	0.454**	0.507**	0.091	-0.126	0.112
IRII (mm hr ⁻¹)	-0.584**	-0.293	0.496**	0.484**	0.058	-0.024	0.316
IRIII (mm hr ⁻¹)	-0.626**	-0.323	0.479**	0.392*	0.009	-0.083	0.405*
Cumulative infiltration	-0.443**	-0.212	0.496**	0.503**	0.069	-0.084	0.245

Note. IR: infiltration rate.

*Significant correlation at the 0.05 level.

**Significant correlation at the 0.01 level.

the minute biomass at this stage is unable to supply excessive root systems, which could only create a small number of spatial voids that can serve as matrix flow pathways, and thus, the soil infiltration rate is also very small. In contrast to grasses, forbs are a more resource-conservative functional group with thicker and deeper-rooted species, and it is difficult for them to settle in a compacted soil surface (Brown & Bugg, 2001). With increasing restoration time, the soil properties improve further, the community structure tends to increase its complexity, and the species richness increases gradually. It has been shown that species richness correlates positively with soil organic carbon and root biomass (Cong et al., 2014; Fischer et al., 2015), which enhance the soil infiltration capacity through promoting the formation

of macropores that strengthen the stability of the soil aggregate (Péres et al., 2013; Ruiz-Colmenero, Bienes, Eldridge, & Marques, 2013).

The present study suggests that soil infiltration rates and cumulative infiltration were positively correlated to the forb functional group ratio (Figure 6m–p); however, a negative relationship with the proportion of grass functional group was found. Archer, Quinton, and Hess (2002) and Fischer et al. (2014) reported that grasses suppress hydraulic conductivity, which is in agreement with the present study. These results could also be explained by the root properties of different functional groups. The soil infiltration capacity is greatly determined by the composition of plant functional groups. Wu et al. (2017) reported that root channels play an important role in the matrix

infiltration ability; the greater infiltration rates of leguminous grasslands and mixed sown grasslands compared with gramineous grasslands could be ascribed to the greater root channels of legumes. At our study site, grasses with fibrous root systems are the first to settle down; fibrous roots usually grow densely in clumps at the soil surface layer and block the soil void, delaying water transport (Archer et al., 2002). As restoration progresses, forbs gradually occupy a larger proportion. The root diameter of forbs is thicker than that of grasses (Figure 5); areas with more forbs tend to support a larger root channel area, which would provide convenient transport for water infiltration due to their increasing effect on hydraulic conductivity (Fischer et al., 2014). The increase of soil infiltration capacity indicates that more water could migrate into the soil and less soil nutrition would be lost (Yu, Shi, Wang, Weidorf, & Zhang, 2008), thus providing favourable conditions for plant colonization and natural revegetation.

5 | CONCLUSIONS

The regulation effect of revegetation on the infiltration ability was investigated in both natural restoration grasslands and untrampled grassland. After excluding trampling interference, grass species had a supreme advantage during the early recovery stage as the root channel area of grasses was smaller than that of forb species. Then, as time passed and the plant communities evolved, the species amount and biomass, especially the forbs, increased gradually. Importantly, the species richness, aboveground biomass, and forb functional group ratio were significantly positively correlated to the soil infiltration ability. Moreover, the R3 grassland had higher capillary porosity and lower bulk density than the R1 grassland, both of which could explain the higher soil infiltration capacity of the R3 grassland.

Our results show that revegetation could tremendously contribute to regulating the soil infiltration processes. With increasing amount of species richness and biomass as well as proportion of forbs, the soil properties were improved, and the soil infiltration capacity enhanced accordingly. This study improves our understanding of the hydrology of natural restoration grassland.

ACKNOWLEDGEMENTS

This research was co-funded by the National Natural Science Foundation of China (nos 31600380, 41807158, 31600379, 31670477, and 31570429) and the National Science Foundation for Post-Doctoral of China (nos 2016M592284 and 2018T110722).

ORCID

Dong Wang  <http://orcid.org/0000-0003-4533-1615>

REFERENCES

- Agnese, C., Bagarello, V., Baiamonte, G., & Iovino, M. (2011). Comparing physical quality of forest and pasture soils in a Sicilian watershed. *Soil Science Society of America Journal*, 75, 1958–1970. <https://doi.org/10.2136/sssaj2011.0044>
- Archer, N. A. L., Otten, W., Schmidt, S., Bengough, A. G., Shah, N., & Bonell, M. (2016). Rainfall infiltration and soil hydrological characteristics below ancient forest, planted forest and grassland in a temperate northern climate. *Ecohydrology*, 9, 585–600. <https://doi.org/10.1002/eco.1658>
- Archer, N. A. L., Quinton, J. N., & Hess, T. M. (2002). Below-ground relationships of soil texture, roots and hydraulic conductivity in two-phase mosaic vegetation in South-east Spain. *Journal of Arid Environments*, 52, 535–553. <https://doi.org/10.1006/jare.2002.1011>
- Bagarello, V., Iovino, M., & Elrick, D. (2004). A simplified falling-head technique for rapid determination of field-saturated hydraulic conductivity. *Soil Science Society of America Journal*, 68, 66–73. <https://doi.org/10.2136/sssaj2004.6600>
- Ball, B. C., Cameron, K. C., Di, H. J., & Moore, S. (2012). Effects of trampling of a wet dairy pasture soil on soil porosity and on mitigation of nitrous oxide emissions by a nitrification inhibitor, dicyandiamide. *Soil Use and Management*, 28, 194–201. <https://doi.org/10.1111/j.1475-2743.2012.00389.x>
- Blanco-Canqui, H., Wienhold, B. J., Jin, V. L., Schmer, M. R., & Kibet, L. C. (2017). Long-term tillage impact on soil hydraulic properties. *Soil & Tillage Research*, 170, 38–42. <https://doi.org/10.1016/j.still.2017.03.001>
- Breuer, L., Huisman, J. A., Keller, T., & Frede, H. G. (2006). Impact of a conversion from cropland to grassland on C and N storage and related soil properties: Analysis of a 60-year chronosequence. *Geoderma*, 133, 6–18. <https://doi.org/10.1016/j.still.2017.03.001>
- Brown, C. S., & Bugg, R. L. (2001). Effects of established perennial grasses on introduction of native forbs in California. *Restoration Ecology*, 9, 38–48. <https://doi.org/10.1046/j.1526-100x.2001.009001038.x>
- Clark, E. V., & Zipper, C. E. (2016). Vegetation influences near-surface hydrological characteristics on a surface coal mine in eastern USA. *Catena*, 139, 241–249. <https://doi.org/10.1016/j.catena.2016.01.004>
- Cong, W. F., van Ruijven, J., Mommer, L., De Deyn, G. B., Berendse, F., & Hoffland, E. (2014). Plant species richness promotes soil carbon and nitrogen stocks in grasslands without legumes. *Journal of Ecology*, 102, 1163–1170. <https://doi.org/10.1111/1365-2745.12280>
- Critchley, C. N. R., Chambers, B. J., Fowbert, J. A., Sanderson, R. A., Bhogal, A., & Rose, S. C. (2002). Association between lowland grassland plant communities and soil properties. *Biological Conservation*, 105, 199–215. [https://doi.org/10.1016/S0006-3207\(01\)00183-5](https://doi.org/10.1016/S0006-3207(01)00183-5)
- Dunne, T., Western, D., & Dietrich, W. E. (2011). Effects of cattle trampling on vegetation, infiltration, and erosion in a tropical rangeland. *Journal of Arid Environments*, 75, 58–69. <https://doi.org/10.1016/j.jaridenv.2010.09.001>
- Fields-Johnson, C. W., Zipper, C. E., Burger, J. A., & Evans, D. M. (2012). Forest restoration on steep slopes after coal surface mining in Appalachian USA: Soil grading and seeding effects. *Forest Ecology and Management*, 270, 126–134. <https://doi.org/10.1016/j.foreco.2012.01.018>
- Fischer, C., Roscher, C., Jensen, B., Eisenhauer, N., Baade, J., Attinger, B., ... Hildebrandt, A. (2014). How do earthworms, soil texture and plant composition affect infiltration along an experimental plant diversity gradient in grassland? *PLoS One*, 9, e98987. <https://doi.org/10.1371/journal.pone.0098987>
- Fischer, C., Tischer, J., Roscher, C., Eisenhauer, N., Ravenek, J., Gleixner, G., ... Hildebrandt, A. (2015). Plant species diversity affects infiltration capacity in an experimental grassland through changes in soil properties. *Plant and Soil*, 397, 1–16. <https://doi.org/10.1007/s11104-014-2373-5>
- Frouz, J., Prach, K., Pižl, V., Háníl, L., Starý, J., Tajovský, K., ... Řehounková, K. (2008). Interactions between soil development, vegetation and soil fauna during spontaneous succession in post mining sites. *European Journal of Soil Biology*, 44, 109–121. <https://doi.org/10.1016/j.ejsobi.2007.09.002>
- Gao, L., Shao, M., & Wang, Y. (2012). Spatial scaling of saturated hydraulic conductivity of soils in a small watershed on the Loess Plateau, China. *Journal of Soils and Sediments*, 12, 863–875. <https://doi.org/10.1007/s11368-012-0511-3>
- Herrick, J. E., Van Zee, J. W., Belnap, J., Johansen, J. R., & Remmenga, M. (2010). Fine gravel controls hydrologic and erodibility responses to

- trampling disturbance for coarse-textured soils with weak cyanobacterial crusts. *Catena*, 83, 119–126. <https://doi.org/10.1016/j.catena.2010.08.007>
- Huang, Z., Tian, F. P., Wu, G. L., Liu, Y., & Dang, Z. Q. (2017). Legume grasslands promote precipitation infiltration better than gramineous grasslands in arid regions. *Land Degradation & Development*, 28, 309–316. <https://doi.org/10.1002/ldr.2635>
- Jiménez, C. C., Tejedor, M., Morillas, G., & Neris, J. (2006). Infiltration rate in andisols: Effect of changes in vegetation cover (Tenerife, Spain). *Journal of Soil and Water Conservation*, 61, 153–158.
- Jørgensen, P. R., Hoffmann, M., Kistrup, J. P., Bryde, C., Bossi, R., & Villholth, K. G. (2002). Preferential flow and pesticide transport in a clay-rich till: Field, laboratory, and modeling analysis. *Water Resources Research*, 38, 28–1–28–15. <https://doi.org/10.1029/2001WR000494>
- Kato, H., Onda, Y., Tanaka, Y., & Asano, M. (2009). Field measurement of infiltration rate using an oscillating nozzle rainfall simulator in the cold, semiarid grassland of Mongolia. *Catena*, 76, 173–181. <https://doi.org/10.1016/j.catena.2008.11.003>
- Khodaverdilloo, H., Cheraghabdal, H. K., Bagarello, V., Iovino, M., Asgarzadeh, H., & Dashtaki, S. G. (2017). Ring diameter effects on determination of field-saturated hydraulic conductivity of different loam soils. *Geoderma*, 303, 60–69. <https://doi.org/10.1016/j.geoderma.2017.04.031>
- Lange, B., Lüscher, P., & Germann, P. F. (2009). Significance of tree roots for preferential infiltration in stagnic soils. *Hydrology and Earth System Sciences*, 13, 1809–1821. <https://doi.org/10.5194/hess-13-1809-2009>
- Leung, A. K., Garg, A., Coo, J. L., Ng, C. W. W., & Hau, B. C. H. (2015). Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrological Processes*, 29, 3342–3354. <https://doi.org/10.1002/hyp.10452>
- Li, Y., Gao, Z., & Tang, L. (2016). Soil-plant characteristics in an age sequence of *Coronilla varia* L. plantations along embankments. *Journal of Soil Science and Plant Nutrition*, 16, 187–199. <https://doi.org/10.4067/S0718-95162016005000015>
- Liu, S., & Wang, T. (2014). Aeolian processes and landscape change under human disturbances on the Sonid grassland of inner Mongolian Plateau, Northern China. *Environmental Earth Sciences*, 71, 2399–2407. <https://doi.org/10.1007/s12665-013-2640-4>
- Luna, L., Vignozzi, N., Miralles, I., & Solé-Benet, A. (2018). Organic amendments and mulches modify soil porosity and infiltration in semiarid mine soils. *Land Degradation & Development*, 29, 1019–1030. <https://doi.org/10.1002/ldr.2830>
- Mao, L. L., Li, Y. Z., Hao, W. P., Mei, X. R., Bralts, V. F., Li, H. R., ... Lei, T. W. (2016). An approximate point source method for soil infiltration process measurement. *Geoderma*, 264, 10–16. <https://doi.org/10.1016/j.geoderma.2015.09.011>
- Miao, R., Jiang, D., Musa, A., Zhou, Q., Guo, M., & Wang, Y. (2015). Effectiveness of shrub planting and grazing exclusion on degraded sandy grassland restoration in Horqin sandy land in Inner Mongolia. *Ecological Engineering*, 74, 164–173. <https://doi.org/10.1016/j.ecoleng.2014.10.004>
- Miao, R., Qiu, X., Guo, M., Musa, A., & Jiang, D. (2018). Accuracy of space-for-time substitution for vegetation state prediction following shrub restoration. *Journal of Plant Ecology*, 11, 208–217. <https://doi.org/10.1093/jpe/rtw133>
- Milleret, R., Le Bayon, R. C., Lamy, F., Gobat, J. M., & Boivin, P. (2009). Impact of roots, mycorrhizas and earthworms on soil physical properties as assessed by shrinkage analysis. *Journal of Hydrology*, 373, 499–507. <https://doi.org/10.1016/j.jhydrol.2009.05.013>
- Pérès, G., Cluzeau, D., Menasseri, S., Soussana, J. F., Bessler, H., Engels, C., ... Weisser, W. W. (2013). Mechanisms linking plant community properties to soil aggregate stability in an experimental grassland plant diversity gradient. *Plant and Soil*, 373, 285–299. <https://doi.org/10.1007/s11104-013-1791-0>
- Pickering, C. M., & Growcock, A. J. (2009). Impacts of experimental trampling on tall alpine herbfields and subalpine grasslands in the Australian Alps. *Journal of Environmental Management*, 91, 532–540. <https://doi.org/10.1016/j.jenvman.2009.09.022>
- Pietola, L., Horn, R., & Yli-Halla, M. (2005). Effects of trampling by cattle on the hydraulic and mechanical properties of soil. *Soil & Tillage Research*, 82, 99–108. <https://doi.org/10.1016/j.still.2004.08.004>
- Razzaghi, S., Khodaverdilloo, H., & Dashtaki, S. G. (2016). Effects of long-term wastewater irrigation on soil physical properties and performance of selected infiltration models in a semi-arid region. *Hydrological Sciences Journal*, 61, 1778–1790. <https://doi.org/10.1080/02626667.2015.1051981>
- Rodriguez-Iturbe, I., & Porporato, A. (2005). *Ecohydrology of water-controlled ecosystem: Soil moisture and plant dynamics*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511535727>
- Roundy, B. A., Winkel, V. K., Khalifa, H., & Matthias, A. D. (1992). Soil water availability and temperature dynamics after one-time heavy cattle trampling and land imprinting. *Arid Soil Research and Rehabilitation*, 6, 53–69. <https://doi.org/10.1080/15324989209381296>
- Ruiz-Colmenero, M., Bienes, R., Eldridge, D. J., & Marques, M. J. (2013). Vegetation cover reduces erosion and enhances soil organic carbon in a vineyard in the central Spain. *Catena*, 104, 153–160. <https://doi.org/10.1016/j.catena.2012.11.007>
- Shao, H., Sun, X., Wang, H., Zhang, X., Xiang, Z., Tan, R., ... Qi, J. (2016). A method to the impact assessment of the returning grazing land to grassland project on regional eco-environmental vulnerability. *Environmental Impact Assessment Review*, 56, 155–167. <https://doi.org/10.1016/j.eiar.2015.10.006>
- Stampfli, A., Bloor, J. M. G., Fischer, M., & Zeiter, M. (2018). High land-use intensity exacerbates shifts in grassland vegetation composition after severe experimental drought. *Global Change Biology*, 24, 2021–2034. <https://doi.org/10.1111/gcb.14046>
- Stumpf, L., Pauletto, E. A., & Pinto, L. F. S. (2016). Soil aggregation and root growth of perennial grasses in a constructed clay minesoil. *Soil & Tillage Research*, 161, 71–78. <https://doi.org/10.1016/j.still.2016.03.005>
- Sun, D., Yang, H., Guan, D., Yang, M., Wu, J., Yuan, F., ... Zhang, Y. (2018). The effect of land use change on soil infiltration capacity in China: A meta-analysis. *Science of the Total Environment*, 627, 1394–1401. <https://doi.org/10.1016/j.scitotenv.2018.01.104>
- Taboada, M. A., & Lavado, R. S. (1993). Influence of cattle trampling on soil porosity under alternate dry and ponded conditions. *Soil Use and Management*, 9, 139–143. <https://doi.org/10.1111/j.1475-2743.1993.tb00943.x>
- Thompson, S. E., Harman, C. J., Heine, P., & Katul, G. G. (2010). Vegetation-infiltration relationship across climatic and soil type gradients. *Journal of Geophysical Research*, 115, G02023. <https://doi.org/10.1029/2009JG001134>
- Volaire, F., Barkaoui, K., & Norton, M. (2014). Designing resilient and sustainable grasslands for a drier future: Adaptive strategies, functional traits and biotic interactions. *European Journal of Agronomy*, 52, 81–89. <https://doi.org/10.1016/j.eja.2013.10.002>
- Wang, D., Zhang, B., Zhu, L., Yang, Y., & Li, M. (2018). Soil and vegetation development along a 10-year restoration chronosequence in tailing dams in the Xiaolinling gold region of Central China. *Catena*, 167, 250–256. <https://doi.org/10.1016/j.catena.2018.05.007>
- Weiler, M., & Naef, F. (2003). An experimental tracer study of the role of macropores in infiltration in grassland soils. *Hydrological Processes*, 17, 477–493. <https://doi.org/10.1002/hyp.1136>
- Wu, G. L., Liu, Y., Fang, N. F., Deng, L., & Shi, Z. H. (2016). Soil physical properties response to grassland conversion from cropland on the semi-arid area. *Ecohydrology*, 9, 1471–1479. <https://doi.org/10.1002/eco.1740>
- Wu, G. L., Liu, Y., Yang, Z., Cui, Z., Deng, L., Chang, X. F., & Shi, Z. H. (2017). Root channels to indicate the increase in soil matrix water infiltration capacity of arid reclaimed mine soils. *Journal of Hydrology*, 546, 133–139. <https://doi.org/10.1016/j.jhydrol.2016.12.047>

- Wu, G. L., Yang, Z., Cui, Z., Liu, Y., Fang, N. F., & Shi, Z. H. (2016). Mixed artificial grasslands with more roots improved mine soil infiltration capacity. *Journal of Hydrology*, 535, 54–60. <https://doi.org/10.1016/j.jhydrol.2016.01.059>
- Wu, G. L., Zhang, Z. N., Wang, D., Shi, Z. H., & Zhu, Y. J. (2014). Interactions of soil water content heterogeneity and species diversity patterns in semi-arid steppes on the Loess Plateau of China. *Journal of Hydrology*, 519, 1362–1367. <https://doi.org/10.1016/j.jhydrol.2014.09.012>
- Yan, B., Fang, N. F., Zhang, P. C., & Shi, Z. H. (2013). Impacts of land use change on watershed streamflow and sediment yield: An assessment using hydrologic modelling and partial least squares regression. *Journal of Hydrology*, 484, 26–37. <https://doi.org/10.1016/j.jhydrol.2013.01.008>
- Yang, J. L., & Zhang, G. L. (2011). Water infiltration in urban soils and its effects on the quantity and quality of runoff. *Journal of Soils and Sediments*, 11, 751–761. <https://doi.org/10.1007/s11368-011-0356-1>
- Yu, D. S., Shi, X. Z., Wang, H. J., Weidorf, D. C., & Zhang, X. Y. (2008). Function of soils in regulating rainwater in Southern China: Impacts of land uses and soils. *Pedosphere*, 18, 717–730. [https://doi.org/10.1016/S1002-0160\(08\)60067-3](https://doi.org/10.1016/S1002-0160(08)60067-3)
- Zhao, Y., Wu, P., Zhao, S., & Feng, H. (2013). Variation of soil infiltrability across a 79-year chronosequence of naturally restored grassland on the Loess Plateau, China. *Journal of Hydrology*, 504, 94–103. <https://doi.org/10.1016/j.jhydrol.2013.09.039>

How to cite this article: Su L, Yang Y, Li X, et al. Increasing plant diversity and forb ratio during the revegetation processes of trampled areas and trails enhances soil infiltration. *Land Degrad Dev*. 2018;29:4025–4034. <https://doi.org/10.1002/ldr.3173>